

12. What is the primary function of an air classifier?
 - A) Separate based on size
 - B) Separate lightweight and heavy materials
 - C) Remove moisture from waste
 - D) Compact waste
13. Which screening method is often used for composts?
 - A) Oscillating screen
 - B) Trommel screen
 - C) Bucket screen
 - D) Air classifier
14. Which method is widely used for separating recyclables in Asian countries?
 - A) Optical sorting
 - B) Manual separation
 - C) Air classification
 - D) Floatation
15. What is the primary goal of recycling in waste management?
 - A) Reduce labor costs
 - B) Enhanced purity and consistency
 - C) Generate more waste
 - D) Eliminate non-biodegradable waste
16. What is the purpose of the Municipal Solid Waste (Management and Handling) Rules, 2000?
 - A) Regulate industrial waste disposal
 - B) Establish guidelines for municipal solid waste management
 - C) Manage biomedical waste
 - D) Oversee construction waste
17. Which agency is responsible for enforcing the Hazardous Wastes (Management and Handling) Rules in India?
 - A) Ministry of Health
 - B) Methanol and biogas
 - C) National Green Tribunal
 - D) Ministry of Urban Development
18. What is the primary consideration in planning a solid waste management site?
 - A) Groundwater protection
 - B) Aesthetic design
 - C) Proximity to residential areas
 - D) Availability of open land
19. Which parameter is critical in site selection for a landfill?
 - A) Wind direction
 - B) Topography and soil conditions
 - C) Average rainfall
 - D) Distance from transport hubs
20. Which step comes first in site remediation?
 - A) Applying remedial technique
 - B) Site monitoring
 - C) Site assessment and inspection
 - D) Leachate treatment

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is solid waste and solid waste management? Discuss the concept of waste management strategies, with a relevant case study.	8	L*2	1	1
	b) Provide an overview of municipal collection services available in residential areas. Also, explain the various collection systems based on their operational modes.	8	L1	2	2
2.	a) How are individuals exposed to solid waste? How does it affect their health and the environment?	8	L2	1	1
	b) Provide a schematic overview of the functional elements involved in solid waste management.	8	L1	2	1
3.	a) A landfill site in Mumbai faced challenges due to excessive leachate production. The municipal corporation assigned the case to a newly appointed environmental engineer. What key physical parameters should the engineer assess to address the issue effectively? Provide justification for your choices.	8	L3	1	1
	b) Briefly explain the following terms i) Shredding ii) Reuse	8	L1	2	2,7
Unit – II					
4.	a) As a highway engineer exploring the use of waste plastics in road construction, with size reduction technique would you recommend for preparing a material for blending with bitumen? Provide a justification of your choice.	4+4	L3	3	2

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|-------|---|-----|----|---|---|
| b) | What preprocessing steps are necessary to prepare wastewater sludge with a high calorific value for energy production through incineration? | 8 | L2 | 4 | 2 |
| 5. a) | Highlight on the following: | 4+4 | L1 | 3 | 2 |
| | i) Eddy Current separators ii) Crosscurrent air classifier | 8 | L1 | 4 | 2 |
| b) | Define landfill and explain its working with a neat labeled diagram | 8 | L1 | 4 | 2 |
| 6. a) | Briefly explain the methods of landfilling. | 8 | L1 | 4 | 2 |
| b) | How does a thermal volume reduction system function? What is the role of sand as a bed material in its construction? | 8 | L2 | 3 | 2 |

Unit – III

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|-------|---|---|----|---|------|
| 7. a) | What is the importance and significance of the Municipal Solid Waste Management Act in addressing solid waste management challenges? | 8 | L2 | 5 | 1,12 |
| b) | Considering the harmful effects of improper e-waste management on health and the environment, what scientific strategies can be proposed to address this issue effectively? | 8 | L2 | 5 | 1,12 |
| 8. a) | Considering the critical importance of hospital waste segregation and management, what strategies can be proposed for its scientific handling? Highlight the regulatory guidelines that govern its proper management. | 8 | L2 | 5 | 1,12 |
| b) | How can effective site planning and development contribute to successful solid waste management? Explain. | 8 | L2 | 5 | 1,12 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
